

TAKUYA KUBO

LinkedIn: [linkedin.com/in/takuyakubo](https://www.linkedin.com/in/takuyakubo) | GitHub: github.com/tky-aaa

EDUCATION

Nanyang Technological University (NTU), Singapore — *MSc in Financial Engineering* | Jul 2025 – Present

Selected Coursework: Stochastic Calculus, Time-Series Analysis, Machine Learning, Computational Methods

Carnegie Mellon University, USA — *Exchange Programme in Computational Finance* | May 2026 – Jun 2026

Selected Coursework: Financial Computing, Simulation Methods, Term Structure Theory and Practice, Advanced Risk Management

Kanazawa University, Japan — *Master of Mechanical Engineering* | Apr 2023 – Mar 2025

Kanazawa University, Japan — *Bachelor of Mechanical Engineering* | Apr 2019 – Mar 2023

TECHNICAL SKILLS

Programming Languages: Python, C++, R, Java, Ruby

Quantitative / Data: NumPy, pandas, SciPy, statsmodels, scikit-learn, matplotlib

Development / APIs: Git, pytest, IBKR API

Languages: Japanese (Native), English (Professional Working Proficiency)

QUANTITATIVE PROJECT

Statistical Arbitrage in Japanese Equities | Jul 2026 – Present

- Developing an end-to-end research and execution system for cointegration-based statistical arbitrage on Japanese equities, which serves as a platform for ongoing research into strategy design and performance.
- Designed and implemented the full workflow covering stock screening, cointegration analysis, signal generation, portfolio management, backtesting, and order execution.
- As an initial study, compared Full, PCA, and FF3 screening methodologies and evaluated their impact on out-of-sample performance under controlled backtests.
- Developed a configurable backtesting framework supporting alternative screening, spread construction, signal-generation, and trading configurations.
- Integrated the system with the IBKR API for paper-trading execution and continue to extend both the research and implementation.

PROFESSIONAL EXPERIENCE

INVESTTECH, Japan — *Financial Engineering Intern* | Jun 2025 – Sep 2025

- Collected historical Japanese electricity market (JPEX) data from the JEPX website and analysed the data in Python.
- Analysed the distribution of spot market prices across delivery time slots and months to inform the selection of time slots for electricity market bidding.

Government Pension Investment Fund (GPIF), Japan — *Investment Management Department Intern* | Mar 2024

- Evaluated outsourced index equity funds with USD 410M–65B AUM based on benchmark tracking, tracking error, and rolling beta.
- Analysed Japanese large-cap active equity funds using benchmark-relative performance and rolling alpha, and selected two funds from the investment universe.
- Summarised the analysis in a presentation and presented the findings to investment professionals.

RESEARCH EXPERIENCE

Kanazawa University, Japan — *Graduate Researcher, Experimental Materials Science* | Apr 2023 – Mar 2025

- Designed and conducted 63 experimental conditions across seven rubber compounds to investigate how material composition, contact pressure, and surface roughness affect friction and wear characteristics.
- Analysed experimental results and automated data processing and visualisation in Python to compare friction and wear characteristics across different experimental conditions.

Kanazawa University, Japan — *Undergraduate Researcher, Experimental Materials Science* | Apr 2022 – Mar 2023

- Investigated whether adhesion contributes to the friction behaviour of tyre materials on icy surfaces by preparing controlled ice surfaces and measuring adhesion between ice and rubber specimens.